

Grade Eight Science - Static Electricity and Magnetism Assessment Checklist

WRITTEN ASSESSMENT FORMAT:

- Based on Criterion A (See Page 2), Levels 1-2, 3-4, 5-6, and 7-8
- Eight Questions, Two at each level

STATIC ELECTRICITY:

1. Know what static electricity is and what causes it
2. Know the structure of the atom. This includes all the subatomic particles (protons, neutrons and electrons) with their charges; know where the subatomic particles are located as part of the atom
3. Know what electrical forces (attraction and repulsion) are and what it means for an object to be neutral, positive, or negative
4. Know that negative charges can be transferred to other objects (if there is contact) and know that only electrons can move between objects and why
5. Know that like charges repel and unlike charges attract and that charged objects can also be attracted to neutral objects
6. Know what electron affinity is and how to use the Triboelectric Series (electron affinity chart) to determine the electron movement, and final charges on objects. (This chart will be provided for you).
7. Understand how to charge by friction, conduction, induction, and permanent induction. Be able to draw and interpret diagrams (ie. rods, charging materials, attraction and repulsion of objects, electroscopes) and provide explanations of the direction of the movement of electrons, how an object got its charge, and the final charge left on objects using any of these charging methods. You must be able to draw and interpret diagrams of electroscopes like the ones done in class
8. Know what grounding an object does and know how to ground an object
9. Know what polarization of charge means and know when charges will polarize
10. Know what an electric field is

MAGNETISM:

1. Know what magnetism is and what causes it
2. Know what a magnetic field is
3. Know that magnets have both a North and a South pole
4. Know that the like poles of magnets repel each other, while opposite poles attract
5. Know what causes repulsion and attraction in magnets

CRITERION A Rubric: Knowing and Understanding

Achievement Level	Level Descriptor
0	The student does not reach a standard indicated by any of the descriptors below
1-2	The student is able to: ● recall scientific knowledge ● apply scientific knowledge and understanding to suggest solutions to problems set in familiar situations ● apply information to make judgments
3-4	The student is able to: ● state scientific knowledge ● apply scientific knowledge and understanding to solve problems set in familiar situations ● apply information to make scientifically supported judgments.
5-6	The student is able to: ● outline scientific knowledge ● apply scientific knowledge and understanding to solve problems set in familiar situations and suggest solutions to problems set in unfamiliar situations ● interpret information to make scientifically supported judgments
7-8	The student is able to: ● describe scientific knowledge ● apply scientific knowledge and understanding to solve problems set in familiar and unfamiliar situations ● analyze information to make scientifically supported judgments